

Application Performance Monitoring with New Relic

Imagine your e-commerce site slows down during Black Friday. New Relic can tell you which API is slow, which database query is causing bottlenecks, and which container is running out of CPU - all in real-time.



Training Agenda

01

Why New Relic?

Understanding the platform's value proposition and core capabilities

03

Architecture & Setup

How New Relic works and getting started

05

Infrastructure & Logs

Monitoring hosts, containers, and centralized logging

07

Alerts & Incident Management

Setting up proactive monitoring and notifications

02

Observability Basics

The three pillars: metrics, logs, and traces

04

APM Monitoring

Deep dive into application performance monitoring

06

Dashboards & NRQL

Creating custom dashboards and writing queries

08

Use Cases & Demo

Real-world examples and hands-on demonstration

This comprehensive 60-minute training will equip you with practical knowledge to implement and leverage New Relic for your applications.

What is New Relic?

New Relic is a SaaS-based **observability and monitoring platform** that provides comprehensive insights into your applications, infrastructure, and user experience.

APM

Application Performance Monitoring for code-level insights

Infrastructure

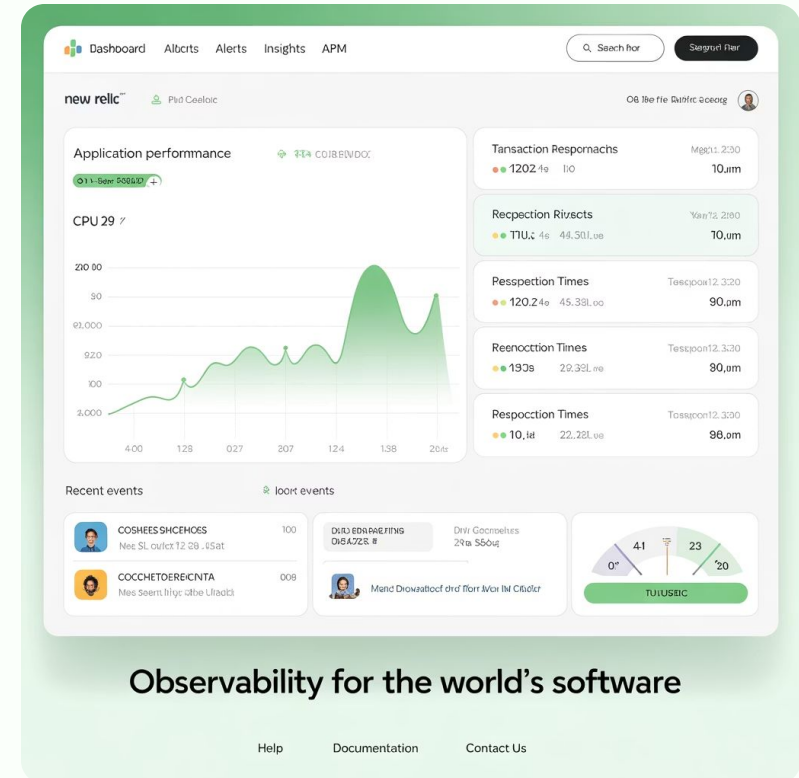
Monitor hosts, containers, and cloud services

Logs

Centralized log management and analysis

Distributed Tracing

End-to-end request flow visualization



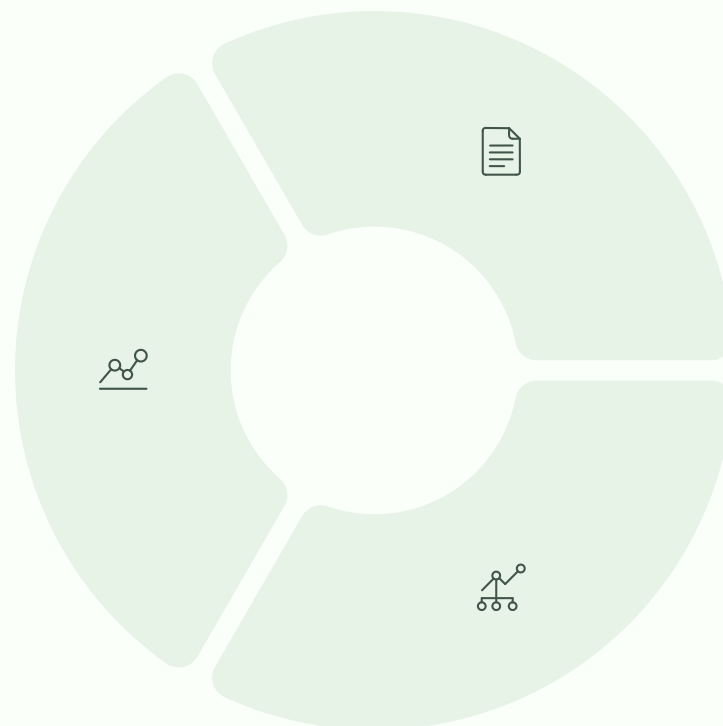
Real-World Impact: Netflix monitors microservices performance while Uber tracks real-time latency across their global platform.

The Three Pillars of Observability

Metrics

Quantitative measurements like CPU usage, memory consumption, and response times

- CPU utilization
- Memory usage
- Response time
- Throughput



Logs

Detailed records of application and system events with timestamps and context

- Application logs
- System logs
- Error messages
- Audit trails

Traces

End-to-end journey of requests through your distributed systems

- Request flow
- Service dependencies
- Bottleneck identification
- Performance correlation

Example: A user request takes 200ms total - metrics show the timing, logs reveal "DB query executed in 180ms", and traces identify the database as the bottleneck.



New Relic Architecture Overview



Application

Your running applications with business logic



Agent

Language-specific agents (Java, Python, Node.js) that instrument your code



New Relic Platform

Telemetry Data Platform that collects and processes all data



Dashboards & Alerts

Visualization and notification systems

The architecture ensures minimal performance impact while providing comprehensive monitoring. Agents automatically collect telemetry data and send it securely to New Relic's platform for analysis and visualization.

Getting Started with New Relic

01

Sign Up

Create a free New Relic account - no credit card required for the free tier

02

Install Agent

Add the appropriate language agent to your application

03

Configure License Key

Set up your unique license key for data collection

04

Deploy & Monitor

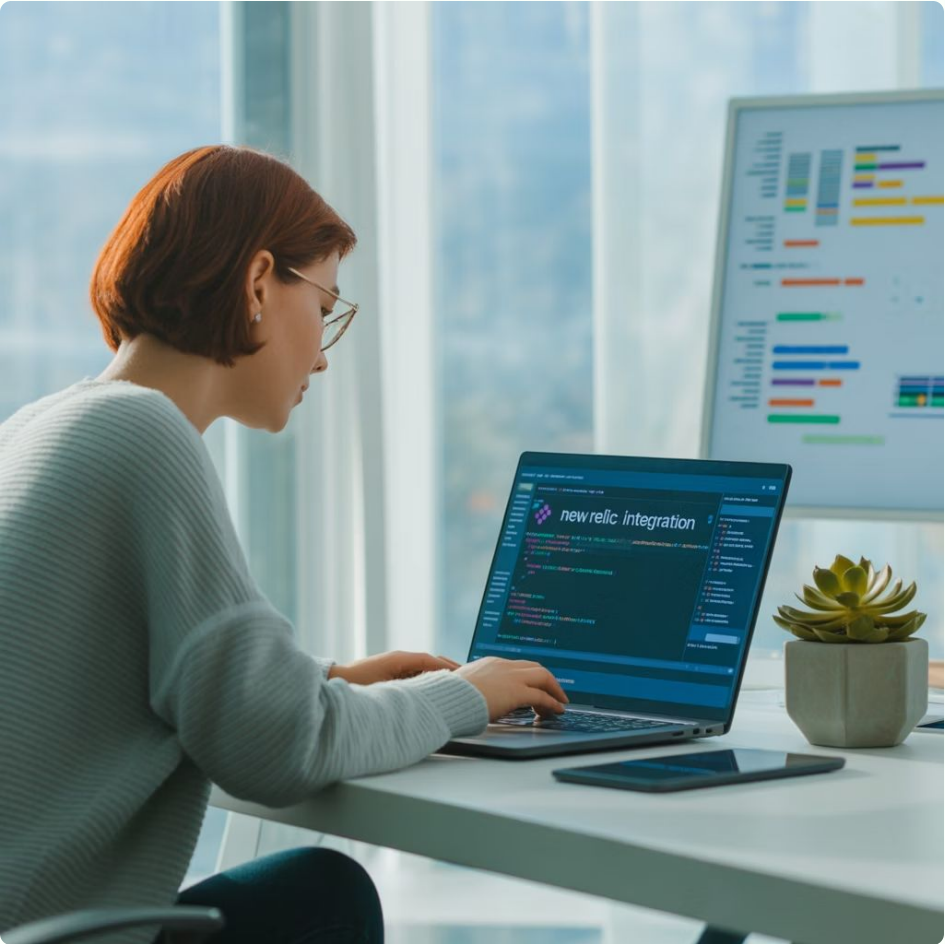
Deploy your instrumented application and start monitoring

Node.js Example Setup

```
npm install newrelic

// Add to top of app.js
require('newrelic')

// Configure in newrelic.js
exports.config = {
  app_name: ['My App'],
  license_key: 'your-license-key'
}
```



APM Deep Dive: Key Metrics

450ms

Response Time

Average request duration for checkout API

1.2K

Throughput (RPM)

Requests per minute during peak hours

8%

Error Rate

Percentage of failed requests (payment service issues)

0.65

Apdex Score

User satisfaction index (poor experience)

These metrics provide immediate insights into application health. An Apdex score below 0.7 indicates user dissatisfaction, while error rates above 5% typically require immediate attention.

E-commerce Example: During Black Friday, the checkout API showed 450ms average response time with 8% error rate, indicating payment service failures that needed immediate resolution.



Infrastructure & Logs Monitoring

Infrastructure Monitoring

-  **Hosts & VMs**
Monitor physical and virtual machines with detailed system metrics
-  **Docker & Kubernetes**
Container-level monitoring with pod health and resource usage
-  **Network & Storage**
Track network performance, disk I/O, and storage utilization

Centralized Logs



Correlate logs with performance metrics and traces for complete visibility into system behavior.

Real Scenario: Kubernetes Pod running out of memory → Infrastructure dashboard shows 90% memory usage → Logs reveal "OOMKilled" event → Immediate scaling action taken.

Dashboards & NRQL Queries

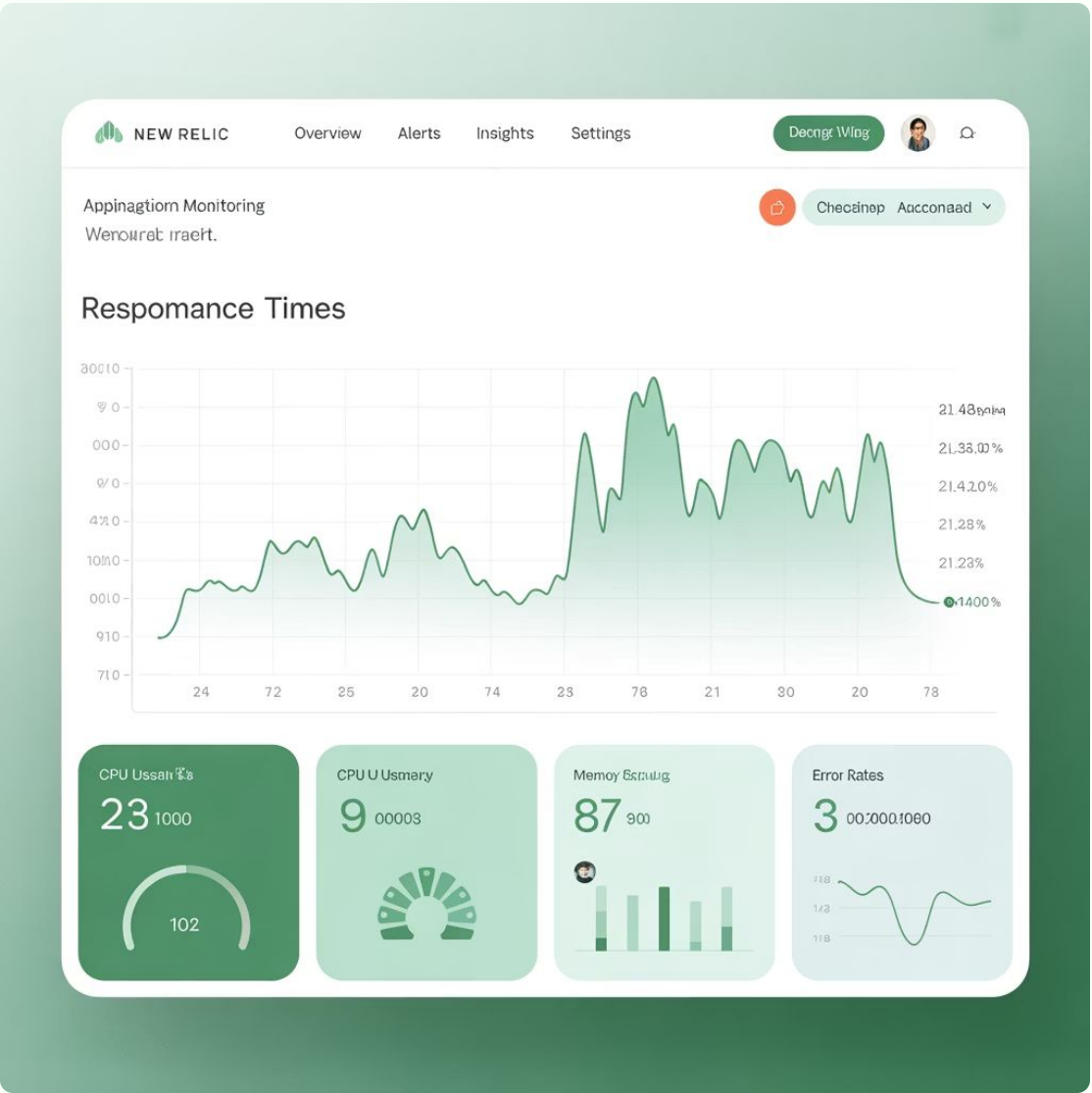
Dashboard Types

- **Pre-built:** APM, Infrastructure, Database dashboards
- **Custom:** Tailored views using NRQL queries
- **Team:** Shared dashboards for collaboration

NRQL Example

```
SELECT average(duration)
FROM Transaction
FACET appName
SINCE 30 minutes ago
```

This query shows average response time per application over the last 30 minutes.



API Latency

Real-time response times across all endpoints

Database Performance

Query execution times and connection pools

User Geography

A smartphone is placed on a white desk next to a small green succulent. The phone screen shows a notification with the text "Critical system update required" and a green button at the bottom that says "Dismiss".

Alerts & Incident Management

01

Define Condition

Set specific thresholds like "CPU usage > 80%" or "Error rate > 5%"

02

Set Duration

Specify how long the condition must persist: "Sustained for 5 minutes"

03

Configure Notifications

Choose channels: Slack, email, PagerDuty, or webhook integrations

04

Incident Response

Automatic ticket creation and escalation workflows



Example Workflow: API error rate exceeds 5% for 10 minutes → Slack alert sent to DevOps team → Auto-ticket created in Jira → On-call engineer notified via PagerDuty

Security & Best Practices



API Key Management

Rotate license keys regularly and use environment-specific keys for different deployment stages



Role-Based Access

Implement granular permissions - developers get read access while admins control configurations



Data Encryption

Encrypt sensitive data in logs and ensure PII is properly masked or excluded



Cost Optimization

Filter unnecessary metrics, set appropriate log retention periods, and monitor data ingestion

Log Security Example

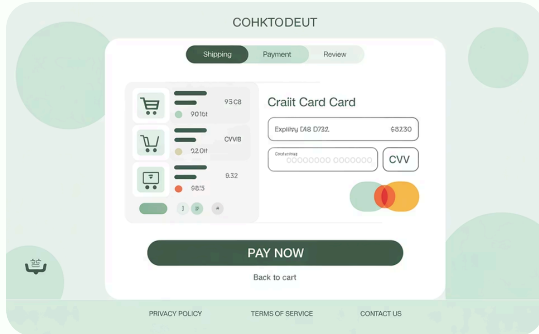
// Good: Hide sensitive data

```
{"user": "admin", "password": "***", "action": "login"}
```

// Bad: Exposing credentials

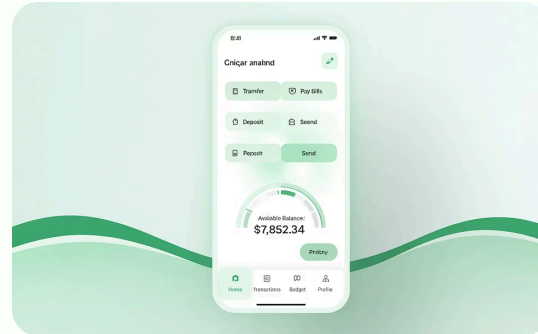
```
{"user": "admin", "password": "secret123", "action": "login"}
```

Real-World Use Cases



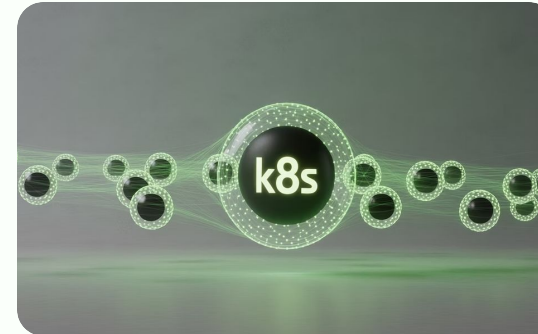
E-commerce

Monitor checkout latency and failed transactions during peak shopping periods like Black Friday



Banking

Track slow database queries in payment systems and ensure transaction processing meets SLA requirements



Kubernetes

Detect pod crashes, monitor container CPU usage, and track resource allocation across clusters



Microservices

Trace request paths across services and identify bottlenecks in distributed architectures

Microservices Trace Example: Frontend → API Gateway (50ms) → Payment Service (800ms) → Database (700ms bottleneck identified)

Training Summary & Next Steps

What We Covered

- New Relic fundamentals and architecture overview
- Application Performance Monitoring (APM) deep dive
- Infrastructure monitoring and centralized logging
- Custom dashboards and NRQL query language
- Alert configuration and incident management
- Security best practices and cost optimization
- Real-world use cases across different industries

Recommended Next Steps

01

Try Free Tier

Set up a demo application with New Relic monitoring

02

Practice NRQL

Experiment with custom queries and data exploration

03

Build Team Dashboard

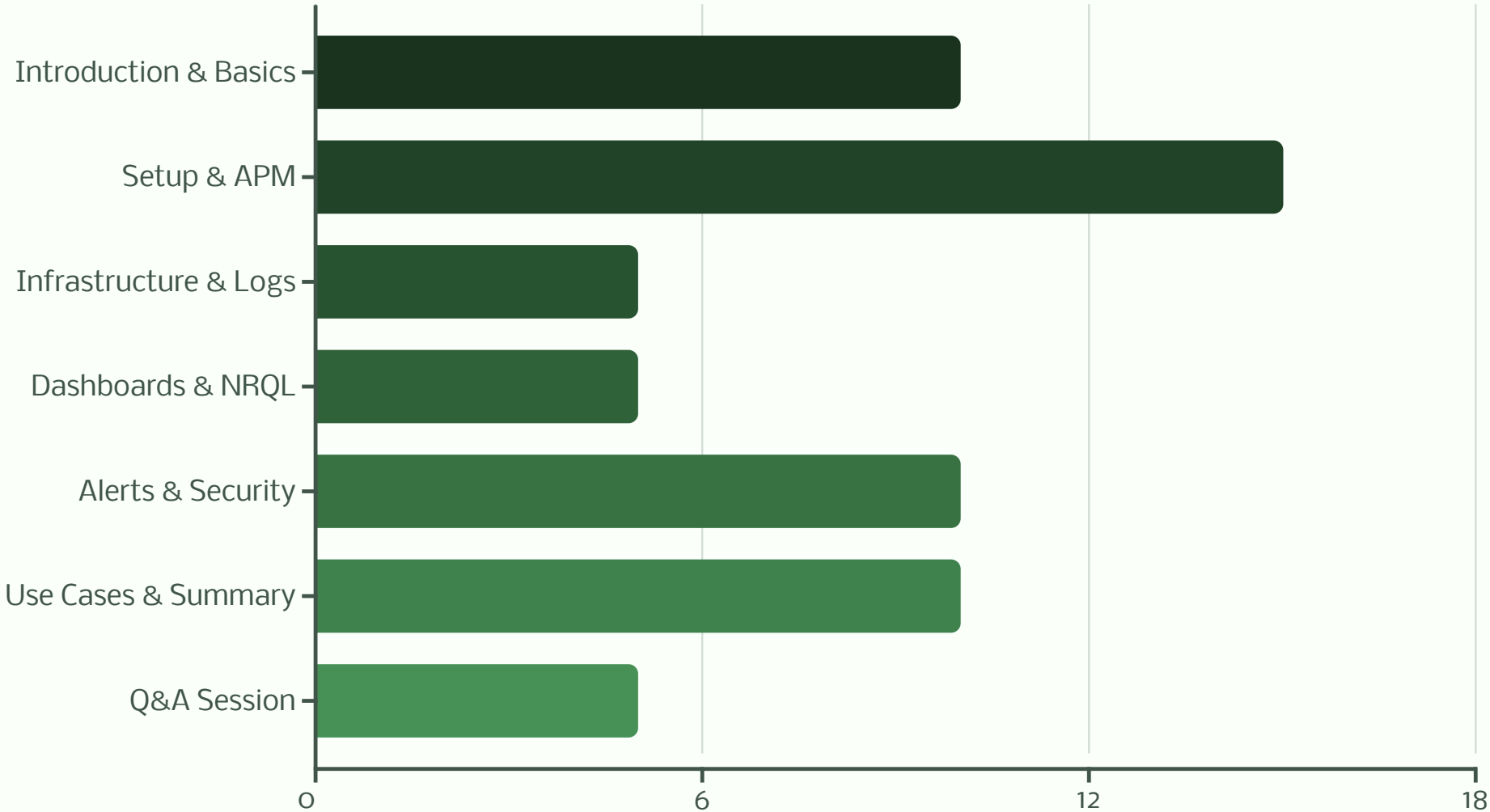
Create shared dashboards for your development team

04

Configure Alerts

Set up proactive monitoring for critical applications

Session Time Breakdown



This structured 60-minute format ensures comprehensive coverage while maintaining engagement through varied content delivery and interactive elements.

Essential Screenshots to Include



New Relic APM Dashboard

Show the main APM overview with response times, throughput, and error rates for a sample application



Infrastructure Monitoring

Display Kubernetes pods monitoring or EC2 instance metrics with CPU, memory, and network usage



NRQL Query Results

Demonstrate a custom NRQL query with results showing application performance data



Slack Alert Example

Show a real Slack notification triggered by New Relic alert with incident details and context

These visual examples will help participants understand the practical application of concepts covered in the training session.

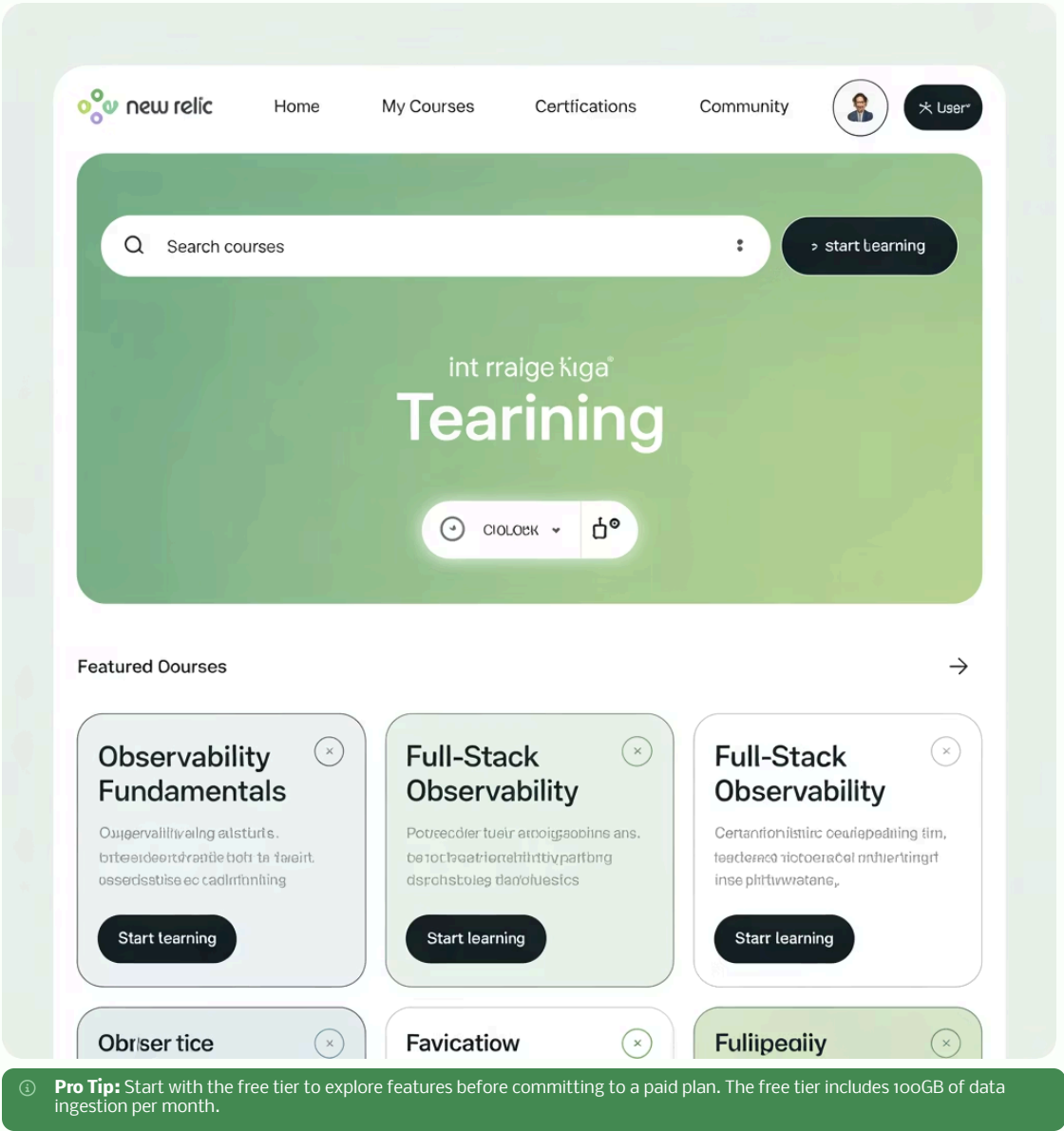
Additional Resources & Documentation

Official Resources

- **New Relic University:** Free online courses and certifications
- **Documentation Hub:** Comprehensive guides for all features
- **Community Forum:** Peer support and best practices sharing
- **GitHub Examples:** Sample applications with New Relic integration

Learning Path

1. Complete New Relic Fundamentals course
2. Practice with sample applications
3. Explore advanced NRQL techniques
4. Implement in production environment



Questions & Discussion

Technical Questions

How does New Relic handle high-volume applications? What's the performance impact of agents?

Implementation Concerns

What are the best practices for rolling out New Relic across multiple teams and environments?

Cost Considerations

How can we optimize data ingestion costs while maintaining comprehensive monitoring coverage?

Thank you for your attention! Let's discuss your specific monitoring challenges and how New Relic can address them.

[Start Your Free Trial](#)

[Schedule Follow-up Session](#)

